

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term Examination, 2021
Department of Materials Science and Engineering

Hum 2127
(Accounting and Economics)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY Three** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY Three** questions from this section in **answer script A**)

- Q 1. (a) State the importance and necessity of accounting in engineering. (10)
(b) What is meant by accounting information? Who are the users of accounting information? (10)
(c) What do you mean by accounting cycle? (05)
(d) Distinguish between journal and ledger. (10)
- Q 2. Mahim open his law office on July 1, 2020. During the first month of operations, the following (15+20) transactions occurred:

2020, July-1	Mahim invested Tk 300,000 in cash in the law practice.
2020, July-2	Paid Tk 8,000 for July rent on office space in cash.
2020, July-3	Purchased equipment in cash Tk 50,000.
2020, July-4	Performed legal services to clients for cash Tk 15,000.
2020, July-5	Cash deposited into bank Tk 70,000.
2020, July-6	Performed legal services for client on account Tk 20,000.
2020, July-7	Advertisement bill received but not yet paid Tk 5,000.
2020, July-8	Paid advertisement bill by cheque transaction July-7.
2020, July-9	Purchase supplies on account Tk 7,000.
2020, July-10	Utility bill paid by cheque Tk 2,500.

- Required: (a) Prepare a Tabular summary of the transactions.
(b) Pass journal entries for the transactions.

- Q 3. From the following information, prepare necessary ledger accounts: (35)
- | | |
|---------------|---|
| 2021, June-1 | Mr. Amin invested Tk 200,000 in cash. |
| 2021, June-2 | Paid office rent for the month Tk 12,000. |
| 2021, June-3 | Purchased office equipment in cash Tk 30,000. |
| 2021, June-5 | Completed an architectural work and billed client Tk 25,000 on account. |
| 2021, June-8 | Received cash Tk 25,000 from client (transaction June-5). |
| 2021, June-10 | Cash deposited into bank Tk 60,000. |

- Q 4. From the book of Nabila Company, the following trial balance has been prepared on 31st (35)
December 2021.

Nabila Company
Trial Balance 31st December, 2021

Account Titles	Debit (Taka)	Credit (Taka)
Capital		500,000
Machine	800000	
Sales		1770,000
Purchase	600000	
Sales return and purchase return	20,000	24500
Stock 1-1-2021	300000	
Advertisement expense	18,000	

Account Titles	Debit (Taka)	Credit (Taka)
Utility expense	4250	
Accounts receivable	450,000	
Accounts payable		250,000
Salaries	167000	
Transportation in	19500	
Allowance for doubtful debts		5250
Bad debts expense	2000	
Rent expense	100,000	
Cash in hand	9000	
Cash in bank	60,000	
	2549750	2549750

The following adjustments are required:

- Closing stock at 31st December 2021 Tk 350000.
- Annual depreciation on machine @5% pa.
- Make allowance for doubtful debts to Tk 5000.

Required:

- Prepare a statement of comprehensive income for the year ended 31st December, 2021.
- Prepare a statement of owner's equity and
- Prepare a statement of financial position as on 31st December, 2021.

Section B

(Answer ANY Three questions from this section in answer script B)

- Q 5. (a) What is managerial economics? What is the role of managerial economics in engineering? (15)
- (b) What makes engineering economics decisions difficult? (08)
- (c) What is meant by demand in economics? What are the determinants of demand? (12)
- Q 6. (a) What do you mean by Price elasticity of demand? (05)
- (b) Can you calculate the Price elasticity of demand on a Particular point of a linear demand curve? How? (use related figure) (15)
- (c) Show, how a consumer get maximum utility under budget constraint, with the help of indifference map and budget line. (15)
- Q 7. (a) What is Production Possibility Curve? (07)
- (b) Define fixed cost and variable cost. In this context, explain the concept of short run cost and long run cost. (08)
- (c) Graphically explain the short run equilibrium of a firm under perfect competition market. (20)
- Q 8. (a) Write down the differences between GNP and GDP. (07)
- (b) Discuss different methods of measuring National Income. (10)
- (c) What are the causes of inflation? How can we control inflation? (18)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term Examination, 2021
Department of Materials Science and Engineering

MSE 2101
(Crystallography)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY Three** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY Three** questions from this section in **answer script A**)

- Q 1. (a) Draw Cubic Closed Packed (CCP) and Hexagonal Closed Packed (HCP) structure and explain why CCP structure is more ductile than HCP structure? (12)
- (b) Draw and describe fluorite and ABO_3 -type cubic perovskite structure. (12)
- (c) Calculate the ideal c/a ratio of HCP crystal and using the relationship, develop an equation for packing density of the HCP crystal. (11)
- Q 2. (a) Why there are no Base centered cubic and Body centered monoclinic Bravais-type lattice? Explain with necessary figures. (12)
- (b) What is Wigner-Seitz cell? Draw a Wigner Seitz cell and explain how to draw this cell? (10)
- (c) Define and draw tetrahedral and octahedral voids. Show that, the ratio of tetrahedral void(r) and atomic radius(R) is 0.225. (13)
- Q 3. (a) What is Miller Indices? Write down the important features of Miller Indices. (10)
- (b) Write down the relationship between inter-planar spacing (d_{hkl}) and Miller Indices (hkl) for Orthorhombic system with appropriate figures. (13)
- (c) What is planar packing density? Draw the following planes and directions in cubic unit cell (111), (101), (001) and $[\bar{3}31]$, $[\bar{1}11]$, $[1\bar{2}1]$. (12)
- Q 4. (a) What is symmetry? What are the differences between trigonal and hexagonal crystal structures, though they possess similar lattice parameters of $a = b \neq c$, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$. (12)
- (b) What is Symmetry element? Describe various Symmetry elements with necessary figures. (12)
- (c) Define Symmetry operation? Describe Inversion and Translation symmetry operation with proper figures. (11)

Section B

(Answer ANY Three questions from this section in **answer script B**)

- Q 5. (a) What do you mean by Reciprocal Lattice Vector? Explain the relationship between Reciprocal Lattice Vector and d-spacing of a real lattice. (10)
- (b) Draw the side view (with Miller indices) of a monoclinic crystal perpendicular to the Y-axis. From this side view, draw a two-dimensional reciprocal lattice and then extend it to a three-dimensional reciprocal unit cell. (15)
- (c) Define zone axis. Find out the zone axis of the planes (211) and (110). (10)
- Q 6. (a) Derive Bragg's law for diffraction, and show that a reflection of any order can be treated as a first order reflection. (11)
- (b) Why does peak broadening occur in X-ray Diffraction patterns? Explain. (12)
- (c) A diffraction pattern of a cubic crystal of lattice parameter $a = 3.16 \text{ \AA}$ is obtained with a monochromatic X-rays beam of wavelength $1.54 \text{ \AA}$. The first line on this pattern was observed to have $\theta = 20.30^\circ$. Determine the interplanar spacing and the Miller Indices of the reflecting plane. (12)
- Q 7. (a) Write down the experimental conditions for different diffraction methods. Describe the Debye-Scherrer method for diffraction. (12)
- (b) How many orders of reflection are possible if the interplanar spacing is $1.506 \text{ \AA}$ and the wavelength is $0.724 \text{ \AA}$? At what angle will they occur? (12)
- (c) Describe the working principle of the Electron Diffraction method using a schematic diagram. (11)
- Q 8. (a) Distinguish between Engineering Projection and Stereographic Projection. (08)
- (b) Draw a stereographic projection of a cubic crystal on (001) plane using a Wulff Net, if the cubic crystal plane $(\bar{1}\bar{1}1)$ shares a zone with the plane $(\bar{1}\bar{1}0)$ and (001), then plot $(\bar{1}\bar{1}1)$ plane on this stereographic projection and calculate the angle between $(\bar{1}\bar{1}1)$ and (001) poles using Tracing Paper and Wulff Net. (15)
- (c) Draw the operations of the following point group symmetry using stereographic projection: (12)
- (i) Vertical Mirror Plane.
 - (ii) Mirror Plane in the plane of the Projection.
 - (iii) Inversion tetrad axis.

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term Examination, 2021
Department of Materials Science and Engineering
MSE 2103
(Thermodynamics of Materials)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Classify the following thermodynamic systems (10)
- i. A dry cell battery
 - ii. An electric magnetic inside a toy
 - iii. A steam boiler heated by gas
 - iv. A can of pepsi
 - v. A eutectic alloy in turbine blade at 1200 rpm.
- (b) "The first law of thermodynamics deals with the conservation of energy while the second law deals with the degradation of energy" – Explain. (12)
- (c) Define state functions. The relationships between three variables x , y , and z can be represented as $z = x^4y + 2xy^2$. Find the coefficient relations of this function and determine whether or not these functions are state functions. (13)
- Q 2. (a) Show that, in an open unary system, chemical potential is identical to molar Gibbs free energy. (08)
- (b) The volume of a system initially at 250K and 1 atm was changed isothermally from 0.3 m³ to 0.5 m³. The change in internal energy was recorded as 1.45kJ. Determine the change in entropy of the system during this process. (12)
- (c) State and prove that the third law of thermodynamics. Mention one important use of this law. (15)
- Q 3. (a) Prove that, heat absorbed in any reversible isothermal process is the maximum. (08)
- (b) Deduce a relationship between the Gibbs free energy of a system to its entropy and temperature. (12)
- (c) Compute the change in Gibbs free energy of magnesia (MgO) when 1 mole is heated from 298 to 1300K at 1 atm pressure. (15)
- Given, $C_p < MgO > = 48.99 + 3.43 \times 10^{-3}T - 11.43 \times 10^{-5}T^{-2} \frac{J}{mol.k}$;
- $$S_{298}^0 = 26.90 \frac{J}{mol.k}$$
- Q 4. (a) Explain the concept of metastable equilibrium and indicate using suitable example, the importance of metastability in materials science and engineering. (08)
- (b) One mole of nickel initially at 300K and 1 atm pressure is taken through two separate processes (12)
- (i) An isobaric change in temperature to 1000K
 - (ii) An isothermal compression to 1000 atm. Compare the change in enthalpy of nickel for these two processes (estimated).
- Given, $V(300K, 1 atm) = 6.57 \text{ cc/mol}$
- $$\alpha = 40 \times 10^{-6} K^{-1}, \beta = 26 \times 10^{-7} atm^{-1}$$
- $$C_p = 11.17 + 37.78 \times 10^{-3}T + 3.18 \times 10^5 T^{-2} \frac{J}{mol.k}$$

- (c) Derive the conditions for equilibrium of a unary two phase system using a constrained maximum of entropy. (15)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

- Q 5. (a) Define Metastability. Explain this term with respect to Gibbs free energy at constant T and P. (15)
- (b) "If vapor pressure in equilibrium with a condensed phase is measured at a series of temperatures, a logarithm of vapor pressure vs. reciprocal of T plot should be a straight line". Deduce a relationship which satisfies the statement. (20)
- Q 6. (a) Explain the graphical method to determine the partial molar properties from total properties. (10)
- (b) Write down the conditions for equilibrium for multi-component, multi-phase systems. (10)
- (c) Construct the phase diagram for pure silicon using the following information: (15)
- $T_m = 1683\text{K}$, $T_b = 2750\text{K}$, $\Delta H_m = 46.4 \text{ kJ/mole}$, and $\Delta H_v = 297 \text{ kJ/mole}$.
- Q 7. (a) Show that the enthalpy change of mixing of an ideal solution is zero. (07)
- (b) Would an atmosphere containing 15 % CO_2 , 5 % CO and 80 % N_2 oxidize Ni at 1000K? (10)
- Given data: $\text{Ni} + \frac{1}{2} (\text{O}_2) = \text{NiO}$; $K_1 = 5.76 \times 10^7$ at 1000K
- $\text{CO} + \frac{1}{2} (\text{O}_2) = \text{CO}_2$; $K_2 = 1.68 \times 10^{10}$ at 1000K.
- (c) Define fugacity and activity. Deduce the relationships between partial molar properties (ΔS_k), Gibbs free energy (ΔG_k), internal energy (ΔU_k), and volume (ΔV_k) for component k with its activity coefficients. (18)
- Q 8. (a) Explain the prominent features of Ellingham diagram. Explain with at least 2 examples. (20)
- (b) Using the Ellingham diagram in Figure 8(b) to answer the following questions: (15)
- (i) Aluminium can be used to reduce titanium oxide at 1000°C whereas chromium cannot be used, Why?
- (ii) Explain why the CO_2 line is horizontal on the diagram but the CO line has a negative slope.
- (iii) Why is carbon an energetic reducing agent at high temperatures?

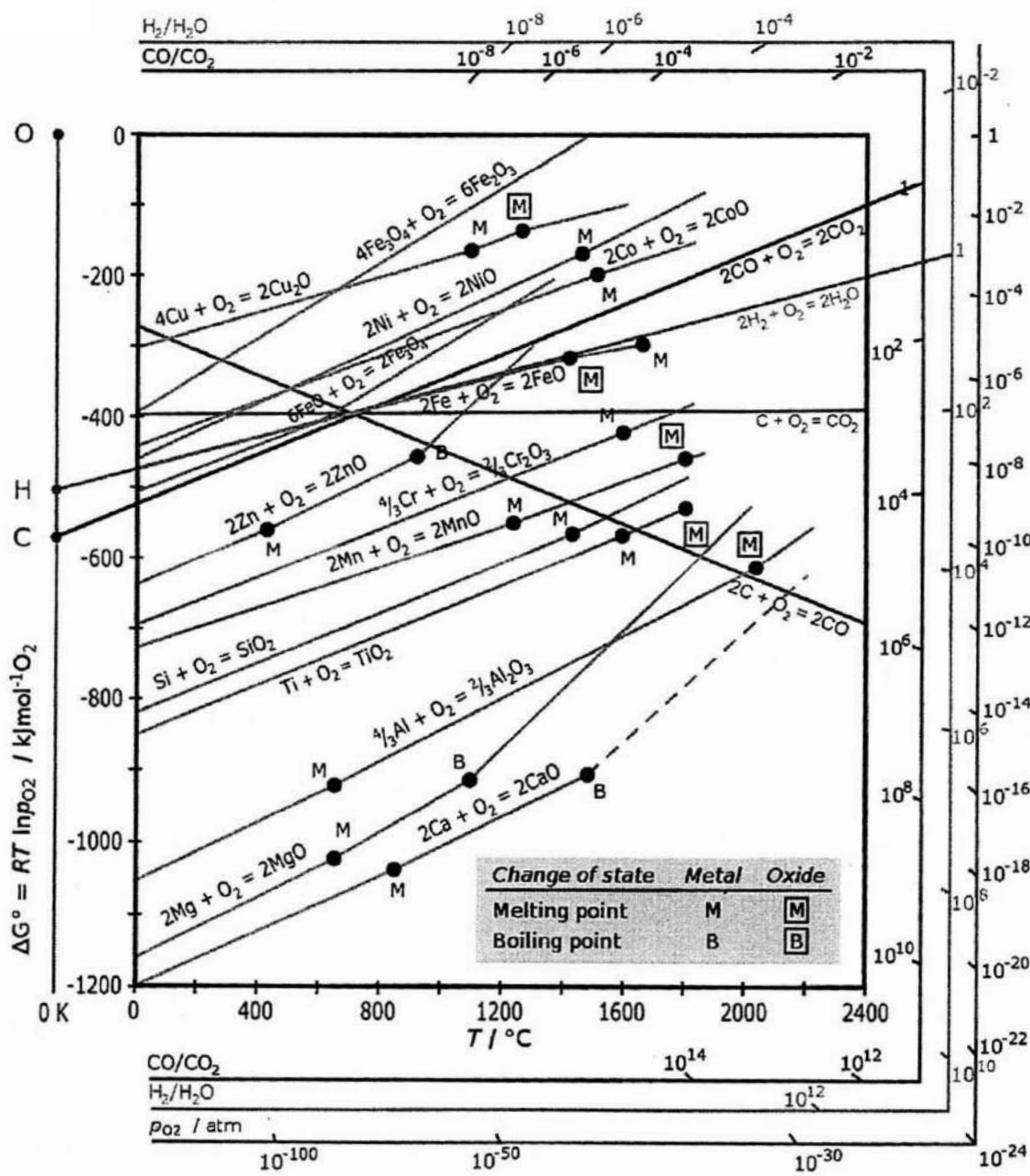


Figure 8(b): The Ellingham Diagram

Khulna University of Engineering & Technology
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Department of Materials Science and Engineering

Math 2127
(Vector Analysis and Linear Algebra)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY Three** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY Three** questions from this section in **answer script A**)

- Q 1. (a) A particle moves along the curve $\vec{r} = (t^3 - 4t)\hat{i} + (t^2 + 4t)\hat{j} + (8t^2 - 3t^3)\hat{k}$, where t is the time. Find the magnitude of the tangential components of its acceleration at $t = 2$. (12)
- (b) What relationship exists between the curvature κ and torsion τ of the space curves $x = t, y = t^2, z = \frac{2}{3}t^3$. (14)
- (c) Write the physical meaning of Gradient, Divergence and Curl. (09)
- Q 2. (a) Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$. (07)
- (b) Find an equation for the tangent plane and normal line to the surface $x^2yz + 3y^2 = 3xz^2 - 8z$ at $P(1, 2, -1)$. Hence find the point in which the normal line meet the plane $x + 3y - 2z = 10$. (14)
- (c) Is the vector $\underline{E} = \frac{r}{r^2}$ irrotational? If so, find the scalar function $f(r)$ such that $\underline{E} = -\underline{\nabla}f(r)$, given that $f(a) = 0$ for $a > 0$. (14)
- Q 3. (a) Prove that $\underline{F} = (y^2 \cos x + z^3)\hat{i} + (2y \sin x - 4)\hat{j} + (3xz^2 + 2)\hat{k}$ is a conservative force field. Also find the work done in moving an object in this field from $(0, 1, -1)$ to $(\frac{\pi}{2}, -1, 2)$. (12)
- (b) To evaluate $\iint_S \underline{A} \cdot \underline{n} \, dS$ use the law of projection, where $\underline{A} = (x + y^2)\hat{i} - 2x\hat{j} + 2yz\hat{k}$ and S is the surface of the plane $2x + y + 2z = 6$ in the first octant. (12)
- (c) By using Green's theorem, evaluate $\oint_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$, where C is the boundary of the region defined by $x = 0, y = 0, x + y = 1$. (11)
- Q 4. (a) Evaluate $\iint_S \underline{F} \cdot \underline{n} \, ds$ by applying divergence theorem, where $\underline{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ and S is the surface of the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$. (11)
- (b) Define: (i) Curvilinear coordinates, (ii) Orthogonal curvilinear coordinates, and (iii) Unit vectors in curvilinear systems. (12)
- (c) Derive an expression for $\underline{\nabla}\phi$ in orthogonal curvilinear coordinates. Hence reduces it to spherical polar coordinates. (12)

Section B

(Answer ANY Three questions from this section in answer script B)

Q 5. (a) Define with example: (i) Minors of a matrix, (ii) Co-factors of a matrix, (iii) Augmented matrix of a system and (iv) Elementary matrix. (12)

(b) Reduce the following matrix A to its echelon form, then to its canonical form, then to its normal form and hence find its rank, where, $A = \begin{bmatrix} 0 & 3 & 1 & -2 \\ 1 & 4 & 3 & -1 \\ 2 & 3 & -4 & -7 \end{bmatrix}$. (10)

(c) Does inverse exist for all matrices? Test whether the following matrix is invertible or not. If possible, find inverse of the following matrix using elementary transform: $\begin{bmatrix} 1 & 20 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$. (13)

Q 6. (a) Choose λ that makes the following system of equations consistent and find the general solution of the system for that λ . (13)

$$x + y - z + t = 2, \quad 2y + 4z + 2t = 3, \quad x + 2y + z + 2t = \lambda.$$

(b) For what values of λ and μ , the following system of equations has (i) no solution, (ii) a unique solution, (iii) infinitely many solutions? (12)

$$x + y + z = 6; \quad x + 2y + 3z = 10; \quad x + 2y + \lambda z = \mu.$$

(c) Find the value of A^2 by using partitioning where, $A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix}$. (10)

Q 7. (a) Define subspace of a vector space. Let, W be a subspace of $\mathbb{R}^4$ generated by the vectors $(1, -2, 5, -3)$, $(2, 3, 1, -4)$ and $(3, 8, -3, -5)$. Find a basis and dimension of W . (13)

(b) Determine whether the vectors $v_1 = (-3, 0, 4)$, $v_2 = (5, -1, 2)$, $v_3 = (1, 1, 3)$ are linearly independent or not in $\mathbb{R}^3$. (10)

(c) Find a spanning Set for the null space of the matrix, $A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$. (12)

Q 8. (a) Find all eigen values and one corresponding eigen vector for the matrix $\begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$. (15)

(b) State Cayley-Hamilton theorem. Verify it for the matrix, $A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$. (12)

(c) Define Quadratic form of a matrix. Express the following quadratic form in the matrix notation $x^T A x$, where A is a symmetric matrix (08)

$$9x_1^2 - x_2^2 + 4x_3^2 + 5x_1x_2 - 8x_1x_3 + x_2x_3.$$

ME 2127
(Solid Mechanics)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) A homogeneous 150 kg bar AB carries a 5 kN load as shown in figure 1(a). The bar is supported by a pin at B, and a 10 mm diameter cable CD. Determine the stress in the cable. (17)
- (b) Determine the force in members BC, BE, and EF of the truss and state if these members are in tension or in compression as shown in figure 1(b). (18)

$$P_1 = 6 \text{ kN}, P_2 = 9 \text{ kN}, P_3 = 12 \text{ kN}$$

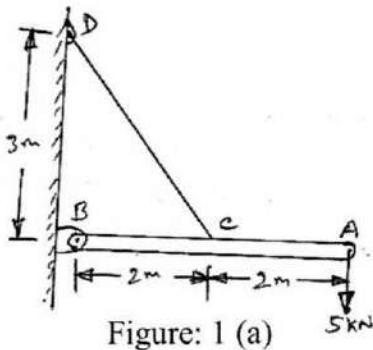


Figure: 1 (a)

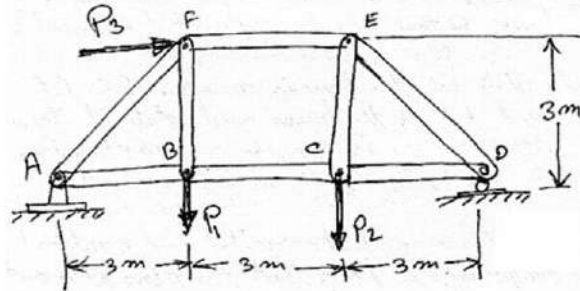


Figure: 1 (b)

2. (a) Determine the horizontal and vertical components of forces that the pins at A and B exert on the frame as shown in figure 2(a). (18)
- (b) In using the bolt cutter shown in figure 2(b), a worker applies two 370 N forces to handles. Determine the magnitude of forces exerted by the cutter on the bolt. (17)

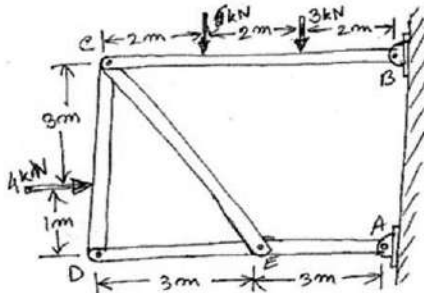


Figure: 2 (a)

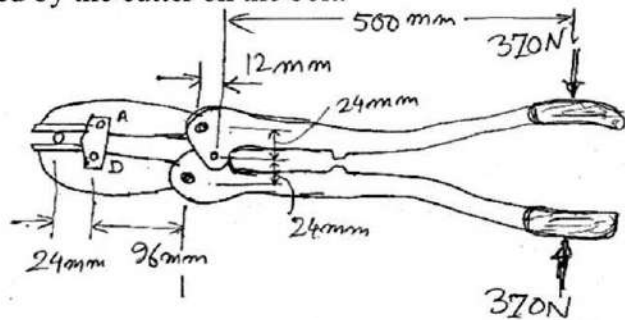


Figure: 2 (b)

3. (a) For the plane area as shown in figure 3(a), determine (i) the first moments with respect to x- and y- axis, and (ii) the location of the centroid. (17)
- (b) Determine by direct integration, the centroid of the area as shown in figure 3 (b). (18)

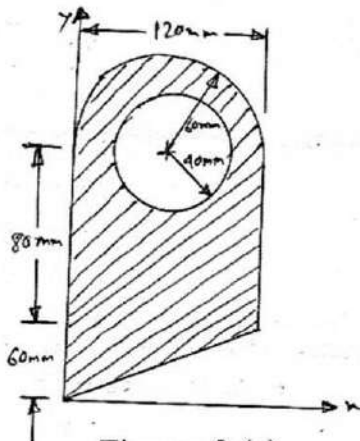


Figure: 3 (a)

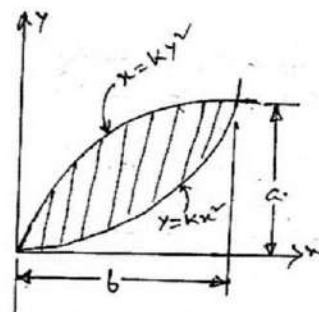


Figure: 3 (b)

4. (a) Determine the moment of inertia of the shaded area as shown in figure 4(a) with respect to y -axis. (17)
- (b) Calculate the moment of inertia and radius of gyration of the shaded area with respect to x -axis as shown in figure 4(b). (18)

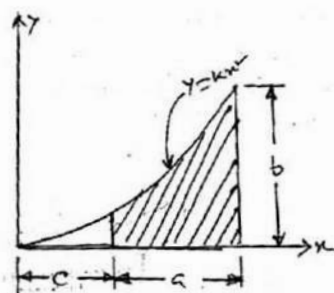


Figure: 4 (a)

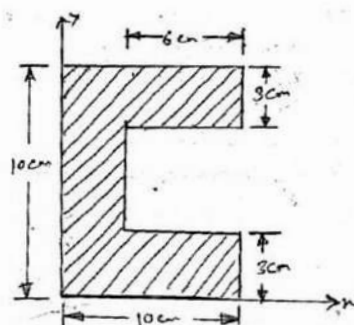


Figure: 4 (b)

Section B

(Answer ANY THREE questions from this section in answer script B)

5. (a) Without writing shear and moment equation, draw the shear and moment diagrams for the beam loaded as shown in figure 5(a). (17)
- (b) The inverted T section of a 5 m simply supported beam has the properties shown in figure 5(b). The beam carries a uniformly distributed load w over the entire length. Determine w if $\sigma_t \leq 40$ MPa and $\sigma_c \leq 80$ MPa. (18)

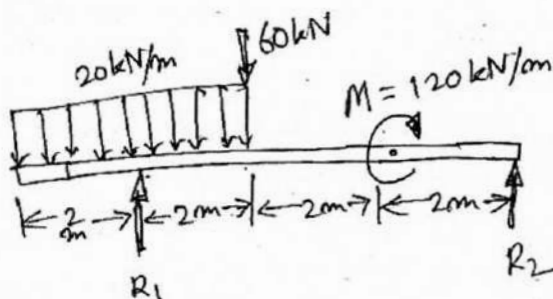


Figure: 5 (a)

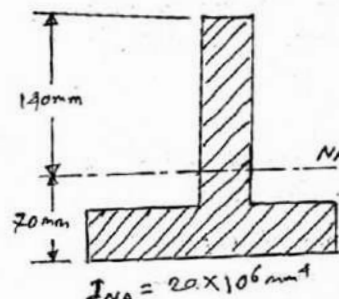


Figure: 5 (b)

6. (a) A rigid horizontal bar of negligible mass is connected to two rods as shown in figure 6(a). If the system is initially stress free, determine the temperature change that will cause a tensile stress of 70 MPa in steel rod. (17)
- (b) A solid compound shaft is made of three different materials and is subjected to two applied torques as shown in figure 6(b). Determine the maximum shearing stress developed in each material and the angle of rotation of the free end of the shaft. (18)
- Use $G_{Al} = 28$ GPa, $G_{St} = 83$ GPa and $G_{Br} = 35$ GPa.

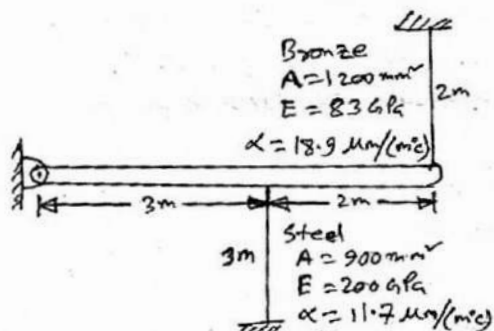


Figure: 6 (a)

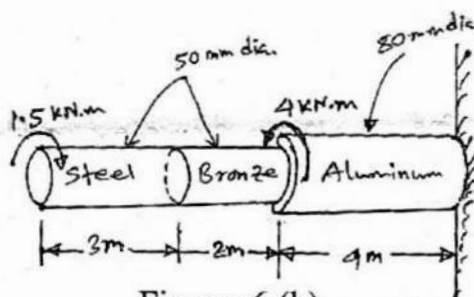


Figure: 6 (b)

7. (a) A rigid bar, hinged at one end, is supported by two identical springs as shown in figure 7 (a). Each spring consists of 30 turns of 15 mm wire having a mean diameter of 200 mm. Compute the maximum shearing stress in the springs. Neglect the mass of the rigid bar. (17)

- (b) The rigid bars AB and CD as shown in figure 7(b) are supported by pin at A and C and the two rods. Determine the maximum force F which can be applied as shown, if its vertical movement is limited to 4 mm. Neglect the weights of all members. (18)

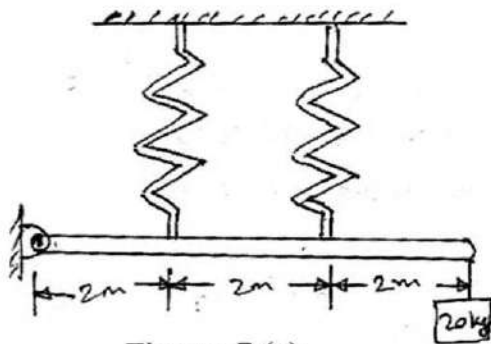


Figure: 7 (a)

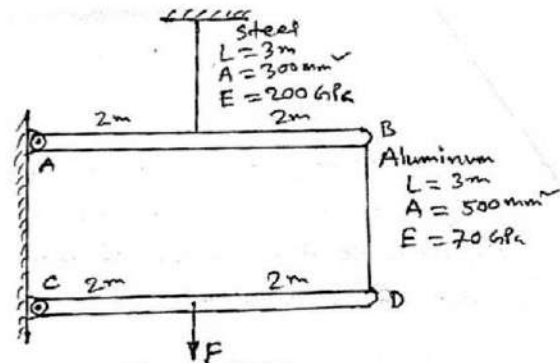


Figure: 7 (b)

8. (a) The bent steel bar shown in figure 8(a) is 200 mm square. Determine the normal stress at A and B. (17)
- (b) For the state of stress as shown in figure 8(b), determine the principle stresses and the maximum shearing stress. Show all results on complete sketches of differential elements. (18)

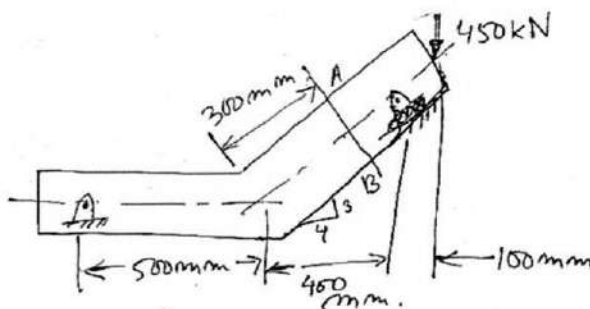


Figure: 8 (a)

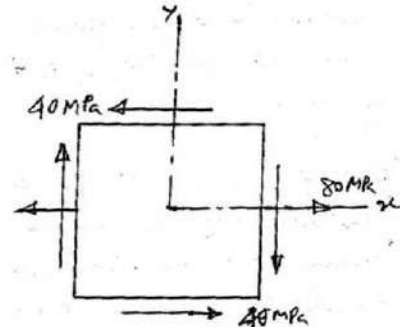


Figure: 8 (b)